In other words, it is a weighted sum, weighted with the probability function *P*.

Alternatively, the average can be defined in an experimental way. Suppose a very large number of identical experiments is made, and the outcomes are recorded. Let's define the probability function in a direct observational manner. We identify $P(\lambda_i)$ as the fraction of observations whose result was λ_i. The definition 4.11 is then identified with the experimental average of the observations. The basic hypothesis of any statistical theory is that if the number of trials is large enough, the mathematical and experimental notions of probability and average will agree. We will not question this hypothesis.

I'll now prove an elegant little theorem that explains the bra-ket notation for averages. Suppose that the normalized state of a quantum system is $|A\rangle$. Expand $|A\rangle$ in the orthonormal basis of eigenvectors of **L**:

$$|A\rangle = \sum_i \alpha_i |\lambda_i\rangle. \qquad (4.12)$$

Just for fun, with no particular agenda in mind, let's compute the quantity $\langle A|\mathbf{L}|A\rangle$. The meaning of this should be clear: First act on $|A\rangle$ with the linear operator **L**.[2] Then, take the inner product of the result with the bra $\langle A|$. Let's do the first step by letting **L** operate on both sides of Eq. 4.12:

$$\mathbf{L}|A\rangle = \sum_i \alpha_i \mathbf{L}|\lambda_i\rangle.$$

Remember that the vectors $|\lambda_i\rangle$ are eigenvectors of **L**. Using the fact that $\mathbf{L}|\lambda_i\rangle = \lambda_i |\lambda_i\rangle$, we can write

$$\mathbf{L}|A\rangle = \sum_i \alpha_i \lambda_i |\lambda_i\rangle.$$